PRACTICE QUESTIONS

Time: 60 minutes

Instructions: Attempt each question before checking the answer. Questions are intentionally mixed across the syllabus. No multiple-choice options are provided.

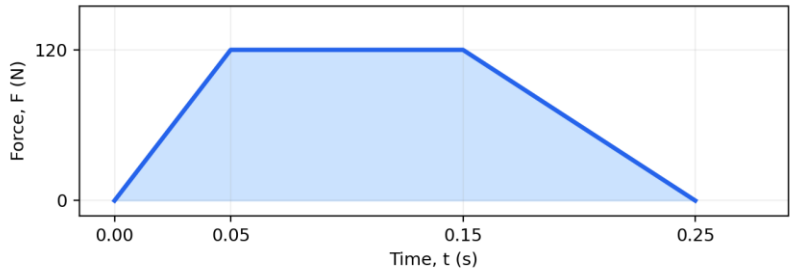
01. A 1.5 kg cart moving at 5 m/s strikes a stationary 2.5 kg cart on a frictionless straight track. The carts couple and move together. Find their common speed immediately after impact.

Answer: 1.88 m/s to the right

02. A particle moves along a straight line with acceleration $a = 3x^2$ m/s², where x is in metres. Its speed is 1 m/s at $x = 0$. Determine its speed at $x = 2$ m.

Answer: 4.12 m/s

03. A horizontal impact force follows the trapezoidal force–time history shown. Determine the total impulse delivered during the 0.25 s interval.

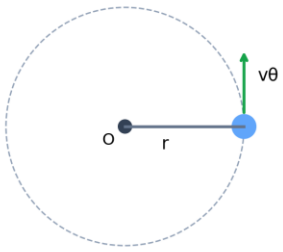


Answer: 24 N·s

04. A 60 kg crate is pulled up a 25° incline at a constant speed of 1.5 m/s. The coefficient of kinetic friction is 0.15 and the pulling cable is parallel to the incline. Determine the cable power.

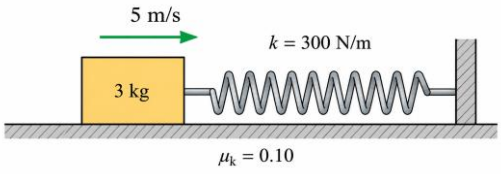
Answer: 0.493 kW

05. A 2 kg puck moves on a smooth horizontal plane under a force directed toward fixed point O. At radius 0.9 m, its transverse speed is 2.5 m/s. What is its transverse speed when its radius becomes 0.6 m?



Answer: 3.75 m/s

06. A 3 kg slider moving at 5 m/s meets an initially undeformed spring of stiffness 300 N/m on a horizontal track. The coefficient of kinetic friction during compression is 0.10. Determine the maximum spring compression.



Answer: 0.490 m

07. A 0.8 kg particle has position vector r = (2i − 3j) m and velocity v = (4i + 2j) m/s. Determine its angular momentum about O, including direction.

Answer: +12.8k kg·m²/s

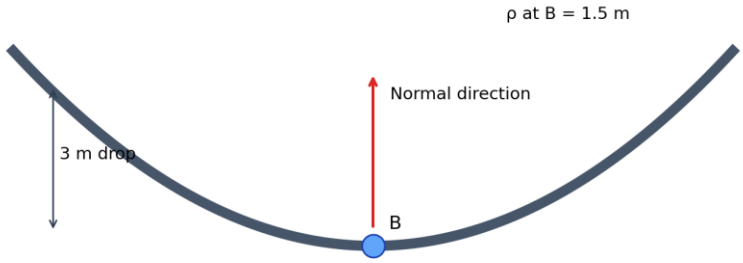
08. A motor-driven wheel has a tangential speed of 12 m/s at the rim, and its speed is increasing at 3 m/s². If the path radius is 24 m, find the magnitude of the rim point's acceleration.

Answer: 6.71 m/s²

09. A 4 kg package slides 5 m down a 20° rough incline from rest. The coefficient of kinetic friction is 0.12. Using work–energy, calculate its speed at the bottom.

Answer: 4.74 m/s

10. A 3 kg bead starts from rest on a smooth curved track and drops vertically through 3 m to its lowest point B. The radius of curvature at B is 1.5 m. Find the normal force exerted by the track at B.



Answer: 147.2 N

11. A constant moment of 8 N·m acts about a fixed point O for 2.5 s on a particle system. Determine the change in angular momentum about O.

Answer: 20 kg·m²/s

12. A 5 kg trolley travels right at 2 m/s. A constant 15-N force acts left for 1.2 s. Determine its final velocity, taking right as positive.

Answer: −1.60 m/s (left)

13. Two smooth balls of masses 1 kg and 2 kg undergo a direct central impact. Initially, the 1 kg ball moves right at 9 m/s, and the 2 kg ball is stationary. With a coefficient of restitution $e = 0.6$, find the velocity of the 2 kg ball immediately after impact.

Answer: 4.80 m/s right

14. A force along the x-axis varies with position according to $F = (12 + 6x^2)$ N, with x in metres. Determine the work done from $x = 1$ m to $x = 3$ m.

Answer: 76 J

15. A 2 kg object initially at rest separates into fragments of 0.5 kg and 1.5 kg during a brief explosion. The lighter fragment travels east at 9 m/s. Find the heavier fragment's velocity, neglecting external impulse.

Answer: 3.00 m/s west

16. A 2 kg collar travels on a horizontal circular guide of radius 0.6 m at constant speed 3 m/s. Determine the magnitude of the required inward resultant force.

Answer: 30 N

17. A block slides down a rough 35° plane with kinetic friction coefficient 0.18. Determine its acceleration along the plane, neglecting air resistance.

Answer: 4.18 m/s² down the plane

18. A particle has polar coordinates $r = 1.5$ m, $\dot{r} = -0.4$ m/s and $\dot{\theta} = 2$ rad/s at a given instant. Determine the magnitude of its velocity.

Answer: 3.03 m/s

19. At an instant, a particle has $r = 1.2$ m, $\dot{r} = 0.5$ m/s, $\ddot{r} = -0.8$ m/s², $\dot{\theta} = 2$ rad/s and $\ddot{\theta} = 1.5$ rad/s². Find the magnitude of its acceleration.

Answer: 6.77 m/s²

20. A 2 kg disk is attached to a central spring on a smooth horizontal plane. Initially, $r_1 = 0.4$ m, $v_{\theta 1} = 3$ m/s and radial velocity is zero; the spring is unstretched and has stiffness 50 N/m. At $r_2 = 0.6$ m, determine (i) transverse speed and (ii) magnitude of radial speed, using conservation of angular momentum and energy.

Answer: (i) 2.00 m/s; (ii) 2.00 m/s